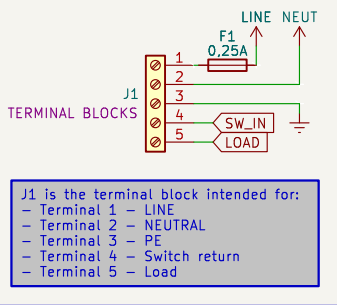


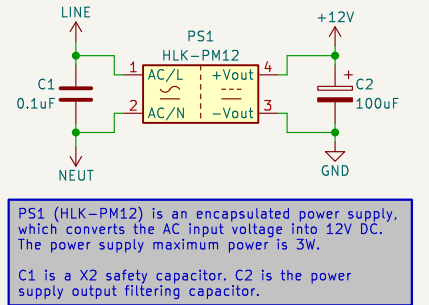
TERMINAL BLOCKS



J1 is the terminal block intended for:

- Terminal 1 - LINE
- Terminal 2 - NEUTRAL
- Terminal 3 - PE
- Terminal 4 - Switch return
- Terminal 5 - Load

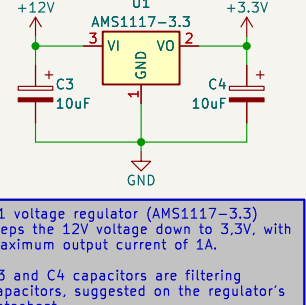
POWER SUPPLY



PS1 (HLK-PM12) is an encapsulated power supply, which converts the AC input voltage into 12V DC. The power supply maximum power is 3W.

C1 is a X2 safety capacitor. C2 is the power supply output filtering capacitor.

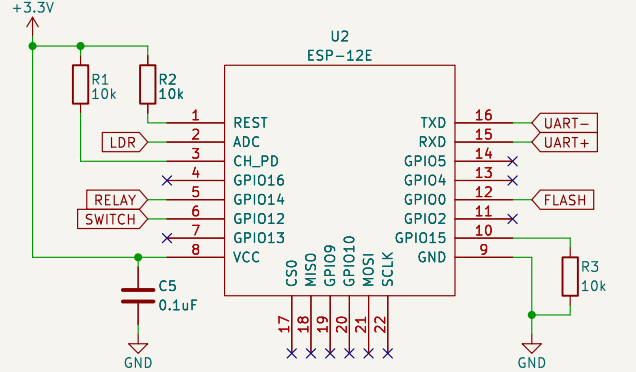
3.3V REGULATOR



U1 voltage regulator (AMS1117-3.3) steps the 12V voltage down to 3.3V, with maximum output current of 1A.

C3 and C4 capacitors are filtering capacitors, suggested on the regulator's datasheet.

MICROCONTROLLER

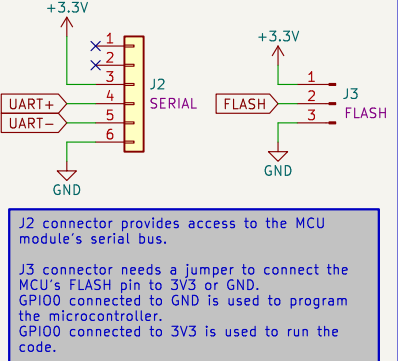


U2 (ESP-12E MODULE) is a microcontroller with WiFi capabilities. It is responsible for all the logic and controll functions of the module.

Some of its pins must be pulled up or down. This is done through the R1, R2 and R3 resistors.

C5 capacitor is a filtering capacitor for the power input.

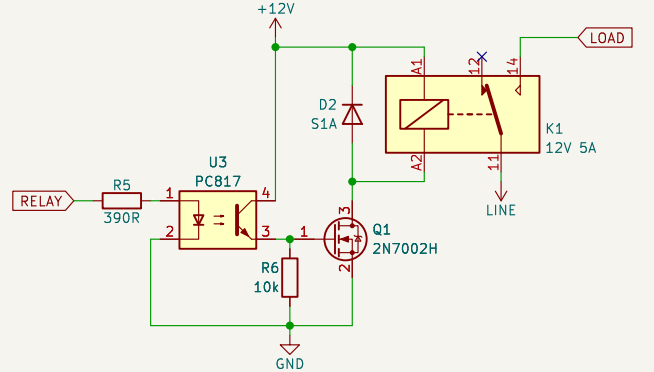
PROGRAMMING CONNECTORS



J2 connector provides access to the MCU module's serial bus.

J3 connector needs a jumper to connect the MCU's FLASH pin to 3V3 or GND. GPIO0 connected to GND is used to program the microcontroller. GPIO0 connected to 3V3 is used to run the code.

RELAY OUTPUT



One of the MCU's digital outputs is used to control the K1 relay through Q1 transistor (2N7002H).

U3 optocoupler (PC817) isolates the output from the power circuit.

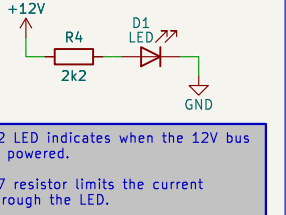
When the output's logic level is LOW, the transistor is not conducting and the relay coil is de-energized.

When the output's logic level is HIGH, the transistor is conducting and the relay coil is energized.

The R5 resistor limits the optocoupler input current. The R6 resistor pull the transistor gate down.

The D1 diode act as a flyback diode. J5 connector goes to the load.

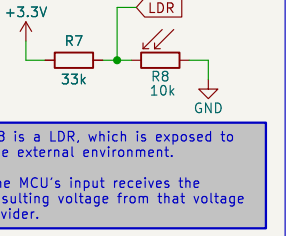
POWER LED



D2 LED indicates when the 12V bus is powered.

R7 resistor limits the current through the LED.

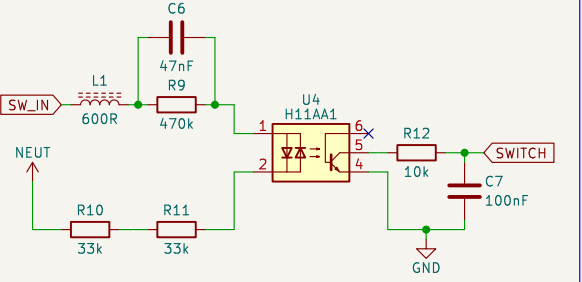
LDR INPUT



R8 is a LDR, which is exposed to the external environment.

The MCU's input receives the resulting voltage from that voltage divider.

SWITCH INPUT



The voltage coming from the switch input is dropped on a capacitive dropper composed by C6, R10 and R11. The impedance is approximately 110 kohms at 60Hz.

L1 is a ferrite bead, intended to reduce high frequency coupling on the switch input.

R9 is the bleeder resistor, which discharge the voltage stored on C6.

U4 (H11AA1) connects the MCU's input to GND when there is a voltage input present.

R12 and C7 compose a RC low-pass filter which cut-off frequency of 160 Hz, approximately.

Alan Carvalho

Sheet: /
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Title: ON-OFF Module X1

Size: A3 Date: 2025-03-22
KiCad E.D.A. kicad-cli 7.0.8

Rev: 2.0
Id: 1/1